

ZHILIN(HADLAY) ZHANG

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Research Interests

My research focuses on **Multimodal Learning** and **Multi-Agent Systems**. I aim to develop robust multi-agent systems that enhance multimodal reasoning capabilities to decompose and solve complex real-world tasks.

Education

The Pennsylvania State University, PA, US

Aug 2026 – May 2031

Ph.D. in Computer Science, advised by Prof. Lu Cheng

New York University, NY, US

Sep 2024 – May 2026

Master of Science in Computer Science, Overall GPA: 3.87/4.0

Core Courses: Machine Learning, Operating System, Algorithmic Machine Learning and Data Science, Deep Learning, Computer Vision, Neuroinformatics

Tongji University, Shanghai, China

Sep 2020 – Jul 2024

Bachelor of Engineering in Computer Science

Core Courses: Data Structure, Software Engineering, Machine Learning, Computer Vision, Natural Language Processing

Honors and Awards: Scholarship for Undergraduate Freshmen (2020-2021); Undergraduate Third-Class Scholarship (2021-2022, 2023-2024)

Publications and Preprints

[†] denotes equal contribution¹. **C** = Conference, **P** = Preprint.

- [C4] Xin Liang[†], **Zhilin Zhang[†]**, Xiang Zhang, Haoran Su, Yiwei Xu, Siqi Sun, Chenyu You. *SlideGen: Collaborative Multimodal Agents for Scientific Slide Generation*. The 34th ACM International Conference on Multimedia (ACM MM 2026).
- [C3] **Zhilin Zhang[†]**, Xiang Zhang[†], Jiaqi Wei, Yiwei Xu, Chenyu You. *PosterGen: Aesthetic-Aware Multi-Modal Paper-to-Poster Generation via Multi-Agent LLMs*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026 (Findings). [paper] [code]
- [C2] **Zhilin Zhang**, Jie Wang, Ruiqi Zhu, Xiaoliang Gong. *Efficient Bilinear Attention-based Fusion for Medical Visual Question Answering*. International Joint Conference on Neural Networks (IJCNN) 2025. [paper]
- [C1] **Zhilin Zhang**, Fangyu Wu. *Enhanced Textual Feature Extraction for Visual Question Answering: A Simple Convolutional Approach*. 6th International Conference on Computer Vision and Computational Intelligence 2025. [paper]

Research Experience

Research Intern, Stony Brook University

May 2025– Present

mentored by Prof. Chenyu You

AI4Research, Multi-agents

- Proposed **PosterGen**, the first aesthetic-aware multi-agent framework capable of generating visually appealing conference posters directly from research papers (  **250+**).
- Designed a specialized multi-agent workflow (Parser, Curator, Layout, and Stylist) that mirrors professional human design pipelines and integrates narrative flow, spatial balance, and cohesive typography.
- Achieved performance significantly surpassing SoTAs in VLM-as-Judge aesthetic design metrics, delivering a tool that substantially alleviates the manual design burden for researchers.

Graduate Research Assistant, University College London

Oct 2024– May 2025

mentored by Dr. Yujian Gan

Multi-agents; Text2SQL

- Developed multiple agents with a layered query parsing architecture and managed robust inter-agent communication to accurately interpret natural language queries and provide reliable feedbacks on UK Census flow data.
- Led the generation of intermediate training data for fine-tuning LLMs by implementing two strategies: rewriting basic APIs and directly generating APIs based on metadata.
- Fine-tuned LLMs (e.g., Llama-3.1) and conducted comprehensive testing and evaluations.

¹Google Scholar: <https://scholar.google.com/citations?user=vWTBvyQAAAAJ>

Undergraduate Thesis, Tongji University
mentored by Prof. Xiaoliang Gong (Tongji University)

Sep 2023– Aug 2024
Vision and Language

- Proposed a fusion framework **OMniBAN** by integrating Orthogonality Loss, Intra-modal Multi-head Self-Attention, and Cross-modal Bilinear Attention Networks to efficiently fuse visual (from **BiomedCLIP**) and textual (from **BioBERT**) representations for MedVQA.
- Conducted extensive comparative experiments on VQA-RAD and SLAKE datasets, and demonstrated that OMniBAN achieves competitive accuracy while reducing computational complexity (around **3.8× fewer FLOPs** and **1.5× fewer parameters**) compared to Transformer-based Co-Attention fusion models.

Undergraduate Research Assistant, Tongji University
Mentored by Prof. Zhongpan Zhu (SRIAS)

Jun 2022– Oct 2022
Multi-modal Re-Identification

- **Second inventor** for the patent: “*Intelligent Unmanned Parking Method based on Spatio-Temporal Vehicle Re-Identification*” [Patent Link](#)
- Proposed a solution for smart parking based on vehicle re-identification, spatial-temporal correlation, sensorless payment, and the architecture of the automated parking system.

Teaching Experience

Teaching Assistant (Spring & Fall 2025, Spring 2026)
CS-GY 6233 Intro to Operating Systems (New York University)

Jan 2025– May 2026
C/C++; Operating System

- Guided by Prof. Omar Mansour.

Additional Skills

- **Standard Tests:** TOEFL **110** (R29, L30, S23, W28), GRE **324** (V154, Q170)
- **Programming Languages:** Python, C/C++, Lean, PHP, Shell
- **Tools & Libraries:** Git, NumPy, Scikit-learn, Pytorch, Keras, LLaMA Factory, LangChain, LangGraph

Academic Services

- **Conference Reviewer:** ACML 2025, WACV 2027
- **Journal Reviewer:** Pattern Recognition, EAAI, IEEE Transactions on Visualization and Computer Graphics